8.2 Diffraction & Interference

Question Paper

Course	CIE A Level Physics
Section	8. Superposition
Topic	8.2 Diffraction & Interference
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which of the following is the definition of diffraction?

A. change of direction when waves cross the boundary between one medium and another

B. splitting of white light into colours

C. addition of two coherent waves to produce a stationary wave pattern

D. bending of waves around an obstacle

[1 mark]

Question 2

Two fringes were created on a screen in a double-slit experiment; the distances were too small to measure.

Which of the following would increase the distance between the fringes?

A. increasing the frequency of the light source

B. increasing the distance between the light source and the slits

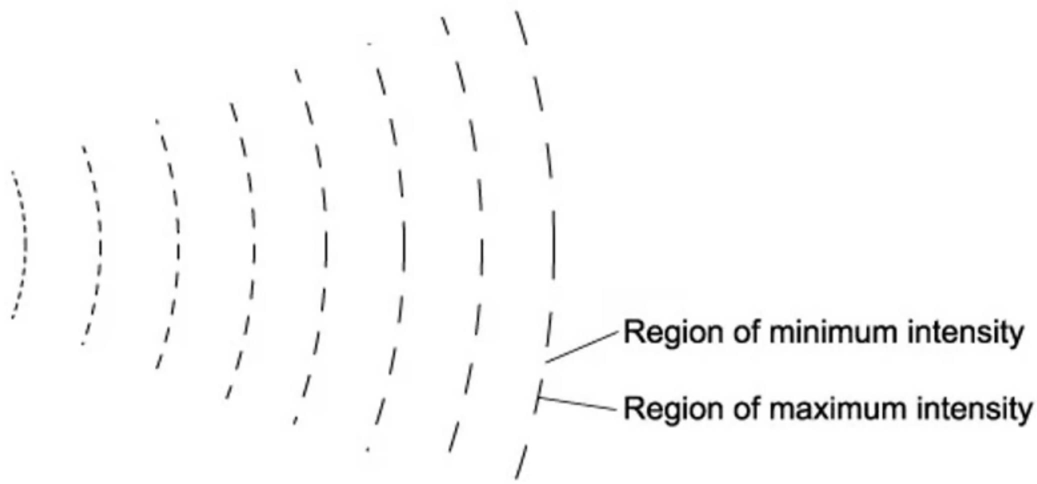
C. increasing the distance between the slits and the screen

D. increasing the distance between the slits

[1 mark]

Question 3

The diagram shows the pattern of waves. The source of the waves was unable to be seen.



Which of the following would cause this pattern?

A. diffraction only

B. interference only

C. coherence only

D. diffraction and interference

[1 mark]

Question 4

An atomic lattice has a spacing of around 10^{-10} m, which electromagnetic wave would cause the most significant diffraction effect?

A. X-ray

B. infra-red

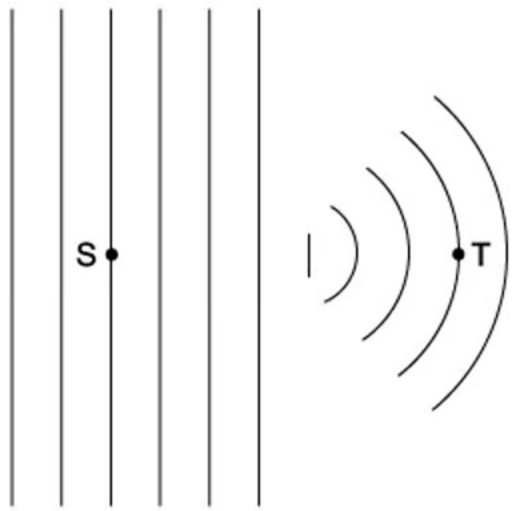
C. microwave

D. ultraviolet

[1 mark]

Question 5

A ripple tank was used to demonstrate plain wavefronts passed through a gap, as shown in the diagram.



Which property would be different at S compared to T?

A. frequency

B. amplitude

C. wavelength

D. velocity

[1 mark]

Question 6

A student investigates the interference of light using two identical green LEDs.

Which statement would explain why the investigation will not work ?

A. the light waves from the sources have a range of wavelengths

B. the light waves from the sources are not monochromatic

C. the light waves from the sources are not coherent

D. the light waves from the sources do not have the same amplitude

[1 mark]

Question 7

A diffraction pattern is observed when monochromatic light was incident on a diffraction grating.

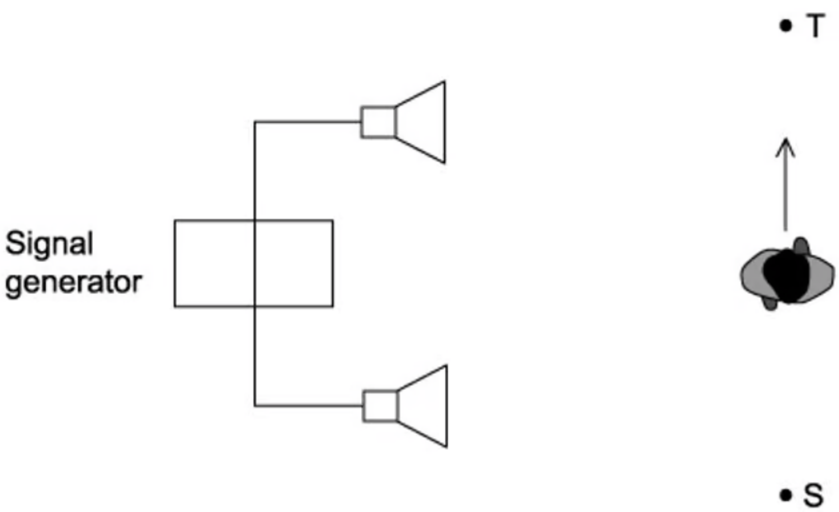
Which row describes the effect of replacing the grating with one that has more lines per metre?

	Number of orders of diffraction visible	Angle between first and second orders of diffraction
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

[1 mark]

Question 8

Two loudspeakers are connected to a signal generator, as shown in the diagram.



As the student walks between S to T she notices that the loudness of the sound increases then decreases repeatedly.

Why does the loudness change?

A. polarisation of the sound waves

B. reflection of the sound waves

C. diffraction of the sound waves

D. interference of the sound waves

[1 mark]

Question 9

When a wave passes an obstruction diffraction can be observed, this effect is greatest when the wavelength and obstruction are similar in size.

When waves are travelling through the air, which of the following would best demonstrate diffraction?

A. radio waves passing a copper wire

B. visible light waves passing a gate post

C. microwaves passing a steel post

D. sound waves passing human hair

[1 mark]

Question 10

A student set up an experiment with a diffraction grating. They put a narrow beam of monochromatic light incident along the normal. They found that the third-order diffracted beams are formed at angles of 45° to the original direction.

Which of the following gives the highest order of diffracted beam produced by this grating ?

A. 6^{th}

B. 5^{th}

C. 4^{th}

D. 3^{rd}

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges

A is incorrect because this definition describes refraction

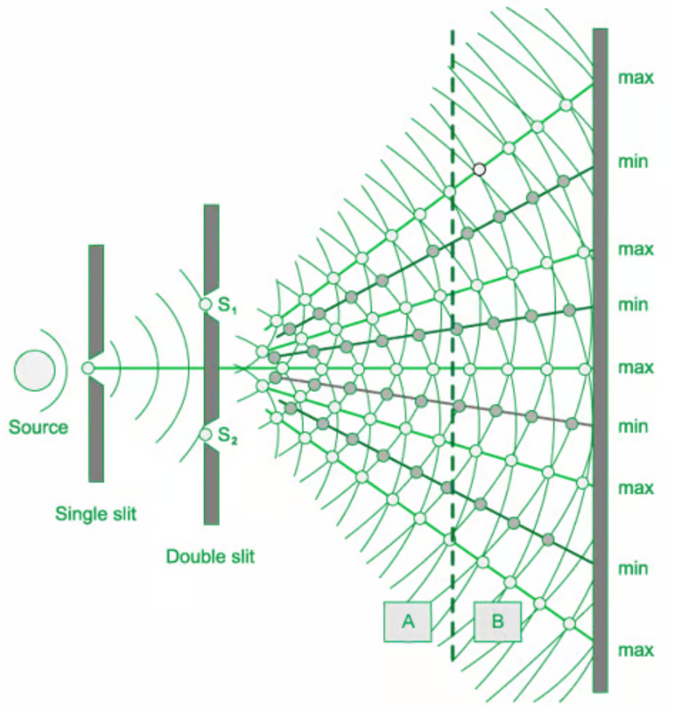
B is incorrect because this definition describes dispersion

C is incorrect because this definition describes superposition

2

The correct answer is **C** because:

- In the diagram below you can see that screen A is closer than screen B



- Fringe separation $\Delta x = \frac{\lambda D}{a}$ where a is the slit separation, λ is the wavelength of light and D the distance from the slits to the screen

A is incorrect because increasing the frequency will decrease the wavelength, as the wavelength decreases this will reduce the diffraction

B is incorrect because increasing the distance between the light source and the slits will change the intensity of the light $I \alpha \frac{1}{a^2}$

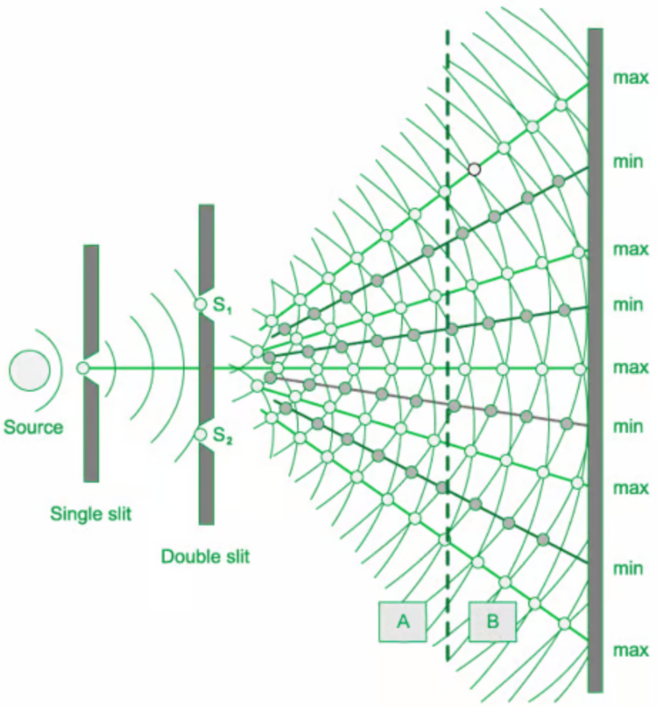
D is incorrect because increasing the distance between the slits would reduce the number of slits in the diffraction grating and reduce the intensity of the maxima lines

3

The correct answer is **D** because:

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- When the light is passed through a double slit two coherent sources of waves are produced that interfere
- Light diffracts from each slit, because the slits are narrow

- These waves overlap and interfere constructively (maximum intensity) and destructively (minimum intensity)



A is incorrect because diffraction through a single slit and would not produce the pattern seen

B is incorrect because the light would need to be diffracted by a single slit to produce the wave pattern seen

C is incorrect because coherence is when the frequency and waveform are identical, and the phase difference is constant. A coherent wave would not produce this wave pattern

4

The correct answer is **A** because:

- Diffraction happens when a wave meets an object or slit
- It is defined as the bending of the waves around corners or through an aperture
- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges
- For the largest diffraction effect to be seen the wavelength needs to be close to

the aperture size
    - X-rays have a wavelength of 10^{-8} to 10^{-13} m

B is incorrect because infra-red waves have a wavelength of 10^{-3} to 10^{-7} m

C is incorrect because microwaves have a wavelength of 10^{-1} to 10^{-3} m

D is incorrect because ultraviolet waves have a wavelength of 10^{-7} to 10^{-8} m

5

The correct answer is **B** because:

- Frequency and wavelength are related to each by the equation : $v = f\lambda$
- The diagram shows the wavefronts at S and T, the only difference between the two points is that the wave has travelled through a slit or gap
- Only the amplitude of the wave changes as some energy is dissipated as it passes through the gap

A is incorrect because the frequency is determined by the source and so remains constant

C is incorrect because the wavelength is constant as the wavefronts are constant in the diagram

D is incorrect because the velocity would remain constant as both wavelength and frequency are constant

6

The correct answer is **C** because:

- In order to observe interference in light waves, the following conditions must be met:
    - The source must be coherent – they must have the same phase
    - The source must be monochromatic – the light must be of a single wavelength
- Interference occurs when light waves are coherent (they have a constant phase difference and the same frequency)
- If the light waves are not coherent, the experiment will not work

- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges
- For the largest diffraction effect to be seen the wavelength needs to be close to the aperture size
  - X-rays have a wavelength of 10^{-8} to 10^{-13} m

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- Interference occurs when light waves are coherent (they have a constant phase difference and the same frequency)
- If the light waves are not coherent, the experiment will not work

A & **B** are incorrect because the LED's are identical, they are monochromatic and have the same wavelength

D is incorrect because amplitude is not a condition for interference

7

The correct answer is **C** because:

- The diffraction grating equation is given by
  - $d \sin \theta = n\lambda$
- Changing the grating to one with more lines per metre causes the slit separation *d*, to decrease
- As the number of order, so diffraction visible decreases the angle between the first order and the second orders of diffraction increases
  - $n = \frac{d \sin \theta}{\lambda}$ as *d* is decreasing
  - $\theta = \sin^{-1}\left(\frac{n\lambda}{d}\right)$, this can only be between 0 and 90˚, and the angle increases

8

The correct answer is **D** because:

- When two or more sound waves from different sources are emitted at the same time, they interact with each other
  - This is called interference
- If the compressions and rarefactions of the two waves line up, they constructively interfere, creating a wave with a higher intensity or loudness
- As the student walks from point S to T, the sound will increase in amplitude due to the constructive interference

A is incorrect because sound waves are longitudinal waves, and longitudinal waves can't be polarised

B is incorrect because reflection of sound waves is an echo and will not increase the amplitude of the sound

C is incorrect because diffraction happens when a wave meets an object or slit

9

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- For very small aperture sizes the majority of the wave is blocked
- For large aperture sizes, the wave passes with very little diffraction, largely at the edges
- For the largest diffraction effect to be seen the wavelength needs to be close to the aperture size
  - Microwaves have a wavelength of 1 m to 10^{-3} m so would be a similar size to the steel post

A is incorrect because radio waves have a wavelength of 10^{-1} to 10^{6} m so would not be diffracted by the copper wire

B is incorrect because visible light has a wavelength of $7 - 4 \times 10^{-7}$ m so would not be diffracted by the gate post

D is incorrect because sound waves have a wavelength of 1.7×10^{-2} m to 17 m so would not be diffracted by the human hair

10

The correct answer is **C** because:

- The diffraction grating equation is given by
  - $d \sin \theta = n\lambda$
  - When $n = 3$ (third order), $\theta = 45°$
- For the highest order of diffracted beam, the angle θ should be less than or equal to 90˚
- Consider the angle of 90°:
  - $d \sin 90 = n\lambda$

- $n = \frac{d \sin 90}{\lambda} = \frac{3 \sin 90}{\sin 45} = 4.24$
- The order should be an integer, so the highest order is $n = 4$